Lecture 19

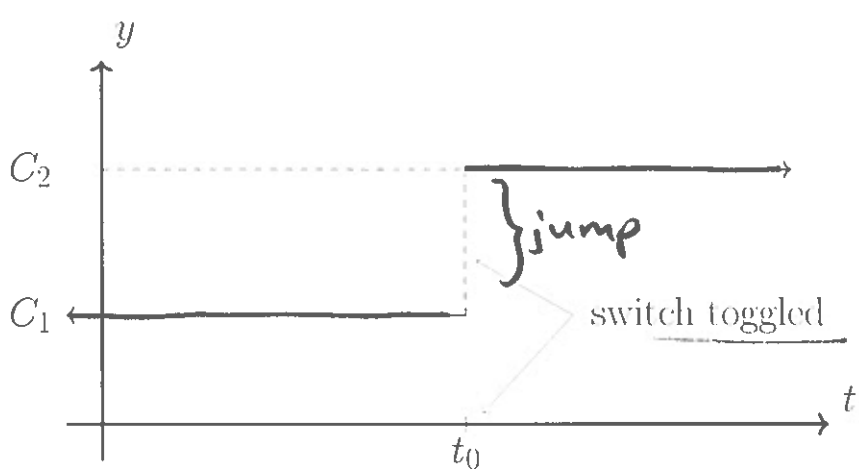
Definition of the Laplace Transform

During our study of differential equations, so far, we have been considering functions which are continuous on their domains or consist of continuous pieces. For instance, we have been considering equations of the form

$$a_n(t)y^{(n)} + \cdots + a_1(t)y' + a_0(t) = f(t)$$

where each of the functions $a_0, a_1, \ldots, a_n$, and f are continuous where we want to consider a solution.

However, this behavior is not always present in applications. Consider the situation of closing a circuit (such as toggling a light switch). There may be current present in the circuit from other lights or devices on the same circuit, but you introduce a new portion of the circuit instantaneously and alter the current when toggling the switch. This type of function cannot be modeled with a continuous curve, but instead must have a jump discontinuity as shown below:



There are many scenarios that can result in this type of behavior: opening and shutting valves to affect fluid mixing and fluid flow, impulses caused by striking objects such as a hammer hitting a nail or a baseball bat striking a baseball, the study of signals and current flow in electronics and circuity, and many others.

We could make use of superposition and solve multiple systems of equations to help deal with the piecewise functions that would arise in such situations, but we will instead employ a new tactic to help handle these types of functions.

Definition: Let $f(t)$ be a function on $[0, \infty)$. The **Laplace transform** of $f(t)$ is a function $F(s)$ defined by

$$\mathscr{L}\{f\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt.$$

new variable

The domain of $F(s)$ consists of all values where the integral converges.

s

NOTE: This integral is an *improper integral* and is evaluated as a limit:

$$\int_0^\infty e^{-st} f(t)\, dt = \underbrace{\lim_{N\to\infty} \int_0^N e^{-st} f(t)\, dt}$$

We will see that this definition enables us to tackle equations where the functions have jump discontinuities or are piecewise defined. Using the Laplace transform will allow us to modify our differential equation into an algebraic equation in the new variable s. Before we can start solving equations, we will first need to build up the theory and techniques around finding Laplace transforms of various functions.

Example: Find the Laplace transform of $f(t) = 1$, $\mathscr{L}\{1\}$, and its domain.

$f(t)=1$

$$\mathscr{L}\{1\} = \int_0^\infty e^{-st}\cdot(1)\,dt = \int_0^\infty e^{-st}\,dt$$

operator

$$= \lim_{N\to\infty}\int_0^N e^{-st}\,dt = \lim_{N\to\infty} -\frac{1}{s}e^{-st}\Big|_0^N$$

$$= \lim_{N\to\infty}\left(-\frac{1}{s}e^{-sN} - \left(-\frac{1}{s}e^0\right)\right) = \lim_{N\to\infty} -\frac{1}{s}\cancel{e^{\underbrace{-sN}_{\to-\infty,\ \underline{s>0}}}}^{\to 0} + \frac{1}{s}$$

$\leftarrow N>0$

$$= \frac{1}{s}$$

So $\mathscr{L}\{1\} = \frac{1}{s}$, $s\neq 0$, $\underline{s>0}$ domain.

Example: Find the Laplace transform of $f(t) = e^{at}$, $\mathscr{L}\{e^{at}\}$, and its domain.

$$\mathscr{L}\{e^{at}\} = \int_0^\infty e^{-st}e^{at}\,dt = \lim_{N\to\infty}\int_0^N e^{-st+at}\,dt$$

$$= \lim_{N\to\infty}\int_0^N e^{-(s-a)t}\,dt = \lim_{N\to\infty} -\frac{1}{s-a}e^{-(s-a)t}\Big|_0^N$$

$$= \lim_{N\to\infty}\left(-\frac{1}{s-a}e^{-(s-a)N} - \left(-\frac{1}{s-a}e^0\right)\right)$$

$$= \lim_{N\to\infty}\left(-\frac{1}{s-a}\cancel{e^{\underbrace{-(s-a)N}_{<0}}}^{\to 0} + \frac{1}{s-a}\right) = \frac{1}{s-a}$$

$s-a>0 \Rightarrow s>a$

$$\mathscr{L}\{e^{at}\} = \frac{1}{s-a}, \quad s>a \leftarrow \text{domain.}$$

Example: Find the Laplace transform of $f(t) = \sin(bt)$, $\mathscr{L}\{\sin(bt)\}$, and its domain.

$$\mathscr{L}\{\sin(bt)\} = \int_0^\infty e^{-st}\sin(bt)\,dt = \lim_{N\to\infty}\int_0^N e^{-st}\sin(bt)\,dt$$

Set $I = \int_0^N e^{-st}\sin(bt)\,dt$

IBP

$\sin(bt)$	$+$	e^{-st}
$b\cos(bt)$	$-$	$-\frac{1}{s}e^{-st}$
$-b^2\sin(bt)$	$\int$	$\frac{1}{s^2}e^{-st}$

$$= \left[-\frac{1}{s}e^{-st}\sin(bt) - \frac{1}{s^2}be^{-st}\cos(bt)\right]_0^N - \int_0^N b^2\sin(bt)\,\frac{1}{s^2}e^{-st}\,dt$$

$$\Rightarrow I = \left[-\frac{1}{s}e^{-st}\sin(bt) - \frac{1}{s^2}be^{-st}\cos(bt)\right]_0^N - \frac{b^2}{s^2}\underbrace{\int_0^N e^{-st}\sin(bt)\,dt}_{I}$$

$$\Rightarrow \left(1 + \frac{b^2}{s^2}\right) I = \left[-\frac{1}{s}e^{-st}\sin(bt) - \frac{1}{s^2}be^{-st}\cos(bt)\right]_0^N$$

$$\frac{s^2+b^2}{s^2} \qquad I = \frac{s^2}{b^2+s^2}\left[\left(-\frac{1}{s}e^{-sN}\sin(bN) - \frac{1}{s^2}be^{-sN}\cos(bN)\right) - \left(-\frac{b}{s^2}\right)\right]$$

So $\mathscr{L}\{\sin(bt)\} = \lim_{N\to\infty}\frac{s^2}{s^2+b^2}\left[\left(\cancel{-\frac{1}{s}e^{-sN}}^{\to 0}\sin(bN) - \cancel{\frac{1}{s^2}be^{-sN}}^{\to 0}\cos(bN)\right) + \frac{b}{s^2}\right]$ $s>0$

$$= \frac{s^2}{s^2+b^2}\left(\frac{b}{s^2}\right) = \frac{b}{s^2+b^2}$$

So $\boxed{\mathscr{L}\{\sin(bt)\} = \frac{b}{s^2+b^2},}$ $s>0$

Example: Find the Laplace transform of $f(t) = t^n$, $\mathcal{L}\{t^n\}$, and its domain, for n a positive integer.

$$\mathcal{L}\{t^n\} = \int_0^\infty e^{-st} t^n \, dt = \lim_{N\to\infty} \int_0^N e^{-st} t^n \, dt$$

IBP

$$\begin{array}{ccc} t^n & + & e^{-st} \\ n t^{n-1} & - & -\frac{1}{s} e^{-st} \end{array}$$

$$\overset{IBP}{=} \lim_{N\to\infty} \left[\left[-\frac{1}{s} e^{-st} t^n \right]_0^N + \frac{1}{s} \int_0^N n t^{n-1} e^{-st} \, dt \right]$$

$$= \lim_{N\to\infty} \left[\left(-\frac{1}{s} \cancel{e^{-sN}}^{\,0} N^n + \frac{1}{s} \cancel{e^0 \cdot 0^n}^{\,0} \right) + \frac{n}{s} \int_0^N e^{-st} t^{n-1} \, dt \right]$$

$s>0$

$$= \lim_{N\to\infty} \frac{n}{s} \int_0^N e^{-st} t^{n-1} \, dt = \frac{n}{s} \underbrace{\int_0^\infty e^{-st} t^{n-1} \, dt}_{\mathcal{L}\{t^{n-1}\}} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

So $\mathcal{L}\{t^n\} = \frac{n}{s} \cdot \mathcal{L}\{t^{n-1}\}$, $n>0$, $s>0$

- If $n=0$, $t^n = 1$, $\mathcal{L}\{t^0\} = \frac{1}{s}$
- If $n>0$, $\mathcal{L}\{t^n\} = \frac{n}{s} \cdot \mathcal{L}\{t^{n-1}\} = \frac{n}{s} \cdot \frac{n-1}{s} \cdot \mathcal{L}\{t^{n-2}\}$

$$= \frac{n}{s} \cdot \frac{n-1}{s} \cdot \ldots \cdot \frac{1}{s} \cdot \mathcal{L}\{1\}$$

$$= \frac{n!}{s^n} \cdot \frac{1}{s} = \frac{n!}{s^{n+1}}$$

$$\Rightarrow \boxed{\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad s>0}$$

$$\mathcal{L}\{t\} = \frac{1!}{s^{1+1}} = \frac{1}{s^2}, \quad \mathcal{L}\{t^2\} = \frac{2!}{s^{2+1}} = \frac{2}{s^3}$$

Theorem (Linearity of the Laplace Transform): Let f_1 and f_2 be functions whose Laplace transforms exists for $s > \alpha$ and c_1 and c_2 be any constants. Then for $s > \alpha$,

$$\mathscr{L}\{c_1 f_1 + c_2 f_2\} = c_1\,\mathscr{L}\{f_1\} + c_2\,\mathscr{L}\{f_2\}$$

$\mathscr{L}\{1\} = \frac{1}{s}$

$\mathscr{L}\{e^{at}\} = \frac{1}{s-a}$

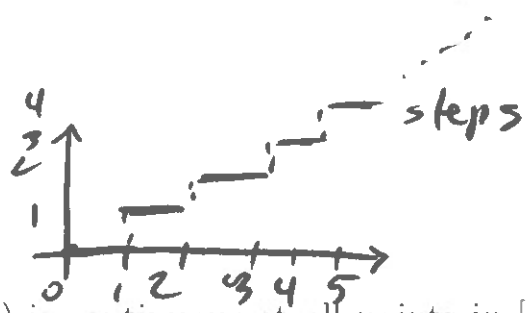
$\mathscr{L}\{\sin(bt)\} = \frac{b}{s^2+b^2}$

Example: Determine $\mathscr{L}\{10 + 7e^{10t} - 5\sin(4t)\}$.

$$= \mathscr{L}\{10\} + \mathscr{L}\{7e^{10t}\} + \mathscr{L}\{-5\sin(4t)\}$$

$$= 10\cdot\mathscr{L}\{1\} + 7\cdot\mathscr{L}\{e^{10t}\} - 5\,\mathscr{L}\{\sin(4t)\}$$

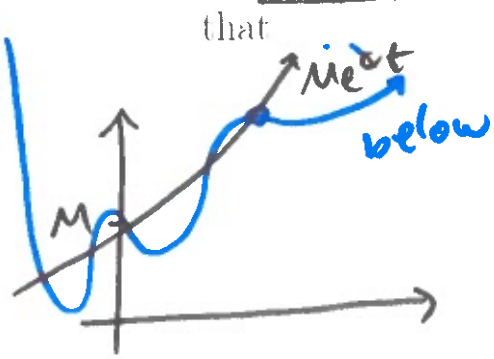
$$= 10\cdot\underbrace{\frac{1}{s}}_{s>0} + 7\cdot\underbrace{\frac{1}{s-10}}_{s>10} - 5\cdot\underbrace{\frac{4}{s^2+4^2}}_{s>0}, \quad s>10$$



Definition: A function $f(t)$ is

- **piecewise continuous** on $[a, b]$ if $f(t)$ is continuous at all points in $[a, b]$ except for a finite number of places where it has a jump discontinuity.
- **piecewise continuous on** $[0, \infty)$ if $f(t)$ is piecewise continuous on $[0, N]$ for all $N > 0$.
- of **exponential order** α if there exists positive constants T and M such that

$$|f(t)| \le Me^{\alpha t} \text{ for all } t \ge T$$



e^t exp order 1

t exp order 1

Theorem (Existence of the Laplace Transform): If $f(t)$ is piecewise continuous on $[0, \infty)$ and of exponential order α then $\mathscr{L}\{f(t)\}$ exists for $s > \alpha$.

Further, if $f(t)$ is piecewise continuous on $[0, \infty)$ and of exponential order α then

$$\mathscr{L}\{f(t)\} = F(s) \to 0 \text{ as } s \to \infty$$

Example: Determine the Laplace transform of

$$f(t) = \begin{cases} t, & 0 < t < 2 \\ 0, & 2 < t < 4 \\ e^{3t}, & t > 4 \end{cases}$$

$$\mathscr{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = \int_0^2 e^{-st} f(t)\,dt + \int_2^4 e^{-st} f(t)\,dt + \int_4^\infty e^{-st} f(t)\,dt$$

$$= \int_0^2 e^{-st} \cdot t\,dt + \cancel{\int_2^4 e^{-st} \cdot 0\,dt}^{\,0} + \int_4^\infty e^{-st} \cdot e^{3t}\,dt$$

$$= \left[-\frac{1}{s} e^{-st} \cdot t + \frac{1}{s^2} e^{-st}\right]\Big|_0^2 + \lim_{N \to \infty} \int_4^N e^{-(s-3)t}\,dt$$

IBP

t	e^{-st}
1	$-\frac{1}{s} e^{-st}$
0	$\frac{1}{s^2} e^{-st}$

$$= \left(-\frac{1}{s} e^{-2s} \cdot 2 + \frac{1}{s^2} e^{-2s}\right) - \left(0 + \frac{1}{s^2}\right) + \lim_{N \to \infty} \left[-\frac{1}{s-3} e^{-(s-3)t}\Big|_4^N\right]$$

$$= e^{-2s}\left(\frac{1}{s^2} - \frac{2}{s}\right) - \frac{1}{s^2} + \lim_{N \to \infty} \left[\cancel{-\frac{1}{s-3} e^{-(s-3)N}}^{\,0} - \left(-\frac{1}{s-3} e^{-4(s-3)}\right)\right] \quad s > 3$$

$$= \boxed{e^{-2s}\left(\frac{1}{s^2} - \frac{2}{s}\right) - \frac{1}{s^2} + e^{-4(s-3)} \cdot \frac{1}{s-3}}$$